Roadmap for the cities

What the Urban-Flow Toolbox is, what we do together, and when

THE GOAL

By April 2028, each pilot city runs its own Urban-Flow Toolbox, installed in its own municipality.

We get there in four steps. In the first one, ENG and the technical partners do the work: your city does not install anything.

3
pilot cities
Valencia · Florence · Tampere

26
tools in the Toolbox
9 system tools + 17 Marketplace tools

April 2028
your own Toolbox, ready
Month 28 of the project (milestone MS6)

What is the Toolbox?

The Urban-Flow Toolbox is a web portal that brings together all the Urban-Flow components. In one place you connect city data, measure what matters, simulate measures before building them, and decide together with citizens.

One web portal for all tools

Each tool has its own page: what it does, who built it and which cities use it.

One login for each city user

Each person sees only what they are allowed to see.

Same base, different tools

The shared base is the same for every city. For each topic (heat, energy, mobility, air) each city chooses the tools it needs.

The Toolbox in one picture

What you use
Local Digital Twin, e-Participation, dashboards, indicators, AI assistant, administration

Marketplace
Tools for one topic: heat, energy, mobility, air. Here the cities differ.

Shared base
Indicators, rules, data and data sharing. The same for every city.

Your city data
Sensors, open data, city systems, maps

Login and permissions
One login. Each person sees only what they are allowed to see.

Data travels from the bottom to the top

1 Collect → 2 Store → 3 Measure → 4 Analyse → 5 Decide

How we get there: four steps

One shared test environment now, and a Toolbox in each city from mid-2027.

1

October – December 2026 · Months 10–12

Setup: We set up the test environment

▼ **We are here**

✕ No installation ✕ No servers

WHAT HAPPENS

- ENG, Tecnalia and the technical partners build one shared test environment for the three cities.
- It is ready in January 2027, when the testing window starts.

WHAT YOUR CITY DOES

- **Nothing to install. No servers.**
- **Your only part: share your requirements and user stories (November) and answer a few technical questions (December).**

Who helps: ENG, Tecnalia and the technical partners

2

January – June 2027 · Months 13–18

Testing window: You test the tools

× No commitment to adopt a tool

× No installation, no servers

× No daily use in operations

WHAT HAPPENS

- The cities try the Marketplace tools and the other components in one shared environment.
- Connect your data and let the tools run their analyses. Try more than one tool for the same topic.

WHAT YOUR CITY DOES

- **Connect your data and play with the Toolbox.**
- **Tell us what you see.**

What you gain. Alternative tools from other partners, and maybe a better option for your final Toolbox.

Who helps: The technical partners support you during the whole testing window

3

June 2027 · Month 18

Feedback: You give feedback and choose

WHAT HAPPENS

- Cities tell us what works and what does not, tool by tool.
- Cities and tool owners go through the feedback together. Owners improve their tools.

WHAT YOUR CITY DOES

- **Give your feedback, tool by tool.**
- **Choose: only the tools that met your expectations go to your own Toolbox.**

What you gain. Your feedback improves the tools and helps the Replicant Cities build on what you learn.

Who helps: Tool owners and the technical partners

4

From mid-2027 to April 2028 · Months 18–28

Implement: You install your own Toolbox

WHAT HAPPENS

- Each city installs its own Toolbox in its own municipality, with the tools that met its expectations.
- There are about 10 months for the local setup and the first implementation.

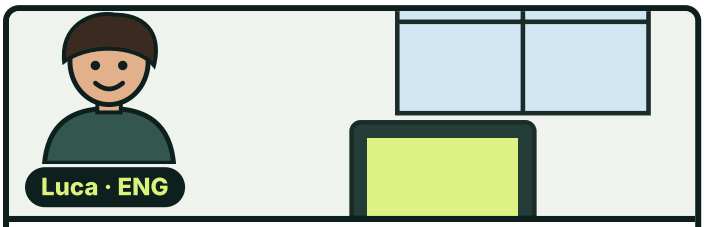
WHAT YOUR CITY DOES

- **Install your own Toolbox, with the technical partners helping you to configure it.**
- **April 2028: your Toolbox is running (milestone MS6).**

Who helps: The technical partners support you in this phase too

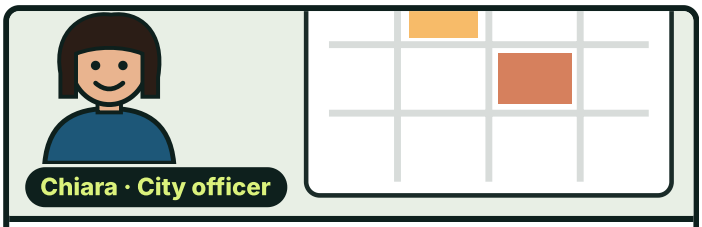
We are flexible. The timeline is flexible. What matters: all the tools are tested, and every city has its own Toolbox in April 2028.

The four steps in four panels



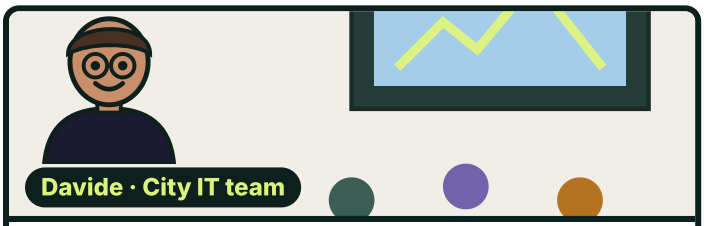
STEP 1 · SETUP

We set up one test environment for all three cities. You install nothing. No servers, no extra work for your IT team.



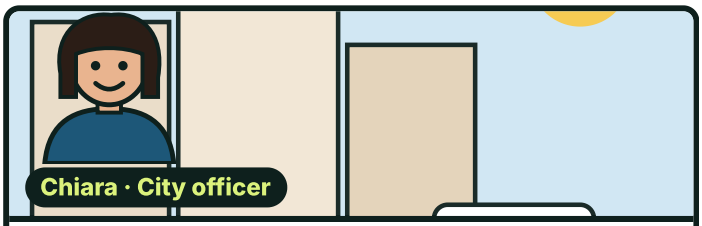
STEP 2 · TESTING WINDOW

From January to June I connect our data and try the tools. I can try two tools for the same topic. It is a test, not a commitment.



STEP 3 · FEEDBACK

In June we say what works and what does not, tool by tool. The owners improve their tools. We keep the ones we like.



STEP 4 · IMPLEMENT

From mid-2027 we install our own Toolbox in the municipality, with help from the technical partners. In April 2028 it is ready.

Which tools will be in your Toolbox?

The Toolbox brings together 26 tools. They are a mix of three kinds: system tools, tools chosen at the General Assembly in Florence, and tools that must be tested to meet the Grant Agreement. **9 + 17 = 26**

9

System tools

The architecture needs them. Without them the Toolbox could not work. They are the same for every city.

+

17

Marketplace tools

Tools the cities asked for at the General Assembly in Florence (28–29 October 2026), plus tools to test, because the Grant Agreement says every tool must be tested at least once.

After the testing window you choose again: only the tools that met your expectations go to your own Toolbox. The system tools are always part of it.

The 9 system tools (same in every city)

e-Participation (Decidim) — ENG
Citizens discuss options and vote.

KPI Manager — ENG
Defines and calculates the indicators.

GeoCacher — ENG
A map-based dashboard of city data.

Data Catalogue (Idra) — ENG
Finds and publishes datasets.

Catalogue of Urban Regeneration Measures — ENG
A library of measures to compare and reuse.

Local Digital Twin viewer — TECNALIA
The city in 3D, with data and simulation results.

AI-based Agent — TECNALIA
Ask questions in plain language.

Chimera and Data Mashup Editor — CEFRIEL, ENG
Prepares and harmonises incoming data.

Minimal Interoperability Mechanisms (MIMs) — OASC
Common rules, so that systems can talk to each other.

The 17 Marketplace tools

How many of these tools each city will test is not decided yet. We will agree it together with each city.

ELECTRA — ETRA
EV bus fleet management

U-NEED — ETRA
Public transport network optimisation & accessibility analysis

EBENTO — ETRA
one stop-shop for energy efficiency tool

INES CLOUD — MUN
Digital urban mobility & parking

MUV CLOUD — MUN
Multimodal mobility & LEZ tracking

UrbanHeatAtlas — TEC
urban heat assessment and planning tool

Climate shelter Router — TEC

Collective DER optimizing tool — TEC

Energy Advising RAG — TEC

TATEBE — VTT
EV charging testbed

Remote Control Center — REM
Automated vehicle control

Kentyou Eye — KENT
Urban hypervisor visualization

EPISODE-CityChem & BenMAP-CE — PV
Air quality/HIA

Tampere.Finland App Integration — BT

PARCAR — ETRA
Dynamic curb-side management

EarthDataInsights — LAT
EDI

Urban Simulator — LAT

Key dates for the cities

When	What happens	Who
November 2026	Requirements and user stories collected together with the cities.	ENG · cities
December 2026	The cities answer the technical questions. The data tools and collaboration platform (V1) are delivered.	Cities · ENG
January 2027	The test environment is ready. The testing window opens.	ENG · Tecnalia
January – June 2027	The cities test the tools and give structured feedback.	Cities · tool owners
June 2027	Feedback and final selection of the tools. First integrated Toolbox (MS4).	Cities · tool owners
June 2027 – April 2028	Local setup and first implementation round in each city.	Cities · technical partners
April 2028	Each pilot city has its own Toolbox (MS6, month 28).	Cities

Questions? Write to info@urban-flow.eu.

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